



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/EMD/374089

Date:

07/08/2023(17:57:39)

Subject:REVISION OF LMI/OGN/OPS/ELEC/023 NEGATIVE PHASE
SEQUENCE ALARMS AT GENERATING STATIONS.

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes are envisaged by COS. Only periodic reviewing is done after two years. No changes in the document.

May please approve.

Submitted by:

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Note No:4

as per clause 7.1 of OGN, appearance of NPS alarm should be intimated to HOD electrical. Accordingly the point may be added to clause 5.4 in LMI.

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Note No:7

Checklist to include the responsibility of filling.

May be routed through all group inahrges for information dissemination to all individuals of EMD and then through me and I/c EMD

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Complied.

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Note No:12

Reviewed the LMI and found in line with OGN.

May process further and approve

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LMI reviewed as per OGN and found in order

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please add revision no. as document is changed

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Revision added.

LMI is re-drafted.

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Issue-2 is to be eliminated every where as revision is done

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Note No:21

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NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION LMI/OGN/OPS/ELEC/023
REVISION: 4, DATE: Aug 2023.

**NEGATIVE PHASE SEQUENCE ALARMS AT
GENERATING STATIONS**

APPROVED FOR

IMPLEMENTATION BY _ _ _ _ _
HOP (SMTTP)

DATE _ _ _ _ _

NTPC LIMITED.

SIMHADRI SUPER THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

LMI(Name) :	NEGATIVE PHASE SEQUENCE ALARMS AT GENERATING STATIONS		
LMI No.	LMI/OGN/OPS/ELEC/023		
ISSUE NO.		Revision : 4	Date :August 2023

	DESIGNATION	SIGNATURE & NAME
Prepared by	DGM (EM)	
Checked by	DGM (EM)	
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Reviewed by	AGM (EM) I/C	
Reviewed by	GM(OPERATION)	
Reviewed by	GM (Maint)	
Reviewed by	GM(O&M)	

NEGATIVE PHASE SEQUENCE ALARMS AT GENERATING STATIONS

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1.0 INTRODUCTION :

Phase unbalance in the generator circuit or in the power system network to which the machine is connected can give rise to Negative Phase Sequence (NPS) currents in the generator. These are responsible for excessive rotor heating with consequential damage to the rotor and possibly to the entire machine.

With the intensive utilization of materials that is being practiced in the design of present-day machines, the margins available are limited and the threat of damage to rotors from cracking or metallurgical degradation, as a result of overheating caused by NPS currents, is significant. Thus, protecting generators against excessive NPS currents is essential. Moreover, a winding fault can be repaired, even stator core damage can be corrected to some extent, but the damage to rotor, which is a single forged unit, is not only irreparable but can also affect the safety of the entire machine/plant.

NPS current protection schemes in NTPC provide for an alarm as well as trip protection, but it should be recommended that there is no provision for back-up to the NPS trip protection in such schemes and it is essential that action is initiated immediately on the appearance of a NPS alarm.

2.0 SCOPE:

Scope of this LMI covers unit 1, 2, 3 & 4 500 MW Generators of Simhadri.

3.0 REFERENCE DOCUMENTS:

Generator manual of 500MW generator of NTPC Simhadri

4.0 SUPERSEDE DOCUMENTS :

NIL

5.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED :

5.1 Origin of NPS currents:

When the power system is healthy, the three phase quantities, both voltage and current, are balanced. That is, they are of the same magnitude but displaced by 120° and rotate in counter clockwise direction (this is known as positive, universally). Under certain abnormal condition (described later) this balance is disturbed and the quantities are no longer symmetrical. For ease of understanding, these unbalanced three phase quantities are split into three symmetrical components, superimposed on one another. These are: -
a) positive phase sequence (the normal three phase components).

- b) negative phase sequence (with clockwise direction of rotation).
- c) Zero phase sequence (with counter clockwise direction of rotation but with nil phase angle difference).

Thus, the negative phase sequence currents circulating in the Generator windings set up a flux that rotates at twice the rotor speed, with the rotor taken as the reference, inducing high levels of eddy currents in rotor iron, slot wedges, damper circuits etc. These currents, in turn, cause overheating of the rotor with subsequent damage.

The rotors in large sized machines installed in modern power stations are with flexibility slits on rotors, i.e., circumferential slits on the pole faces. These slits are to allow adjustment of stresses in the rotor body (between slotted and unslotted portions) while the machine is spinning. But these slits have an adverse effect on the NPS current capability of the rotor in that they force any axially induced rotor currents into the narrow areas of the rotor. This gives rise to high localised heating and cracks initiating from the ends of the slits.

Machines are subjected to a continuous low level of NPS currents, under normal conditions. These originate in the transmission system due to the lack of full transposition of overhead lines and to some extent due to unbalance at the load end.

Higher levels of NPS currents, which can give rise to an Alarm in the Unit Control Room and tripping of the machine if corrective steps are not taken in time, are caused by an abnormal operating condition such as:

1. A tap position in a generator transformer(R, Y, B phases) not being same ("out of step")
2. An open phase in the transmission network
3. An un cleared system fault.
4. One pole of the GCB or 400kV breakers in switchyard failing to close.

5.2 Capability of the Generators protection schemes:

The following is the technical data of NTPC Simhadri 500MW generators:

Rated Data	Turbogenerator
Apparent power	588MVA
Active power	500MW
Current	16.166kA
Voltage	21kV $\pm$ 1.05kV

Speed	50s ⁻¹
Frequency	50Hz
Power factor	0.85(lag)
Inner connection of stator winding	YY
H2 pressure	3.5 bar
Cont. permissible Unbalanced Load	8%
Rated field current for rated output	4040A
Rated field voltage	340V
Type	THDF 115/59

Capability of the generators is universally presented in terms of the following two parameters:

- a. Continuous NPS current withstand capability (SMPP) – 8%
- b. Thermal withstand characteristic of the machine, current Vs time (Represented by I_2^2t). (Enclosed as Annexure)

$I_2 = 8\%$ & I_2^2t not to exceed 10.

The relay has provision for both alarm and tripping. The relay characteristic is set to imitate the thermal characteristic of the machine. The relay also has a memory function that takes into consideration the heating effect due to low level NPS currents and also repetitive short duration NPS currents.

5.3 Settings

All relays have inbuilt algorithms for imitating the characteristics of machine based on the settings set by user. The alarm and trip setting of NPS in protection relays shall be as per approved settings from NTPC Engineering.

The settings employed are:

Alarm: 90% of continuous NPS current capability of the generator. A time delay of about 5 sec is provided in the scheme to avoid spurious alarms during momentary power system faults.

Trip: Tripping doesn't occur as long as NPS currents are not above the continuous rating of the generator and then the tripping time is in accordance with the I_2^2t characteristic that is set slightly below the machine thermal characteristic.

5.4 NPS alarm – action required.

Depending on the origin of NPS currents, it is possible that the appearance of alarm and the tripping of the machine are simultaneous. Tripping may also take place without the appearance of alarm (since alarm is time delayed)

In most cases there will be time available to act after appearance of alarm and before tripping takes place.

The first step to be taken as soon as the alarm comes –

“Reduction of load on the machine until the alarm ceases” and Inform to Electrical Maintenance.

Load reduction assists in reduction of magnitude of NPS currents, thus allowing more time for fault identification and rectification.

Upon receipt of any NPS alarm, operation should immediately inform HOD-electrical and should determine if the alarm has been received:

Situation-1: On one machine only

Situation-2: On all machines in a specific group

Situation-3: On all machines in the station

Situation-1:

If an alarm is received on one generator in the station, then the following may be checked:

1. Check whether all the phases of the GT are on the same tap position
2. One point to be noted about the NPS alarms is whether it occurs immediately after synchronization. If so, then it is advisable to trip the unit and investigate the problem before raising the load. This may be done after confirming from the grid that there is no disturbance occurred which could have led to NPS current.
3. Transformer HV connections are healthy.
4. The bus on which the machine is connected and both the buses are coupled together.

Situation-2: If the alarm appears on a group of machines, it should be verified whether these are connected to the same busbar in the switchyard and that the busbar is not coupled with the other busbar. Then step 5.3 can be followed to identify the cause of unbalance in the system.

Situation-3:

If the alarm has appeared on all the machines in the station, it is recommended to check the balance of currents in all the phases of the out going feeders from the switchyard Control Room by taking the ammeter readings.

Whether both the machines are on the same bus of switchyard and the other bus is out of service or the buses are coupled together.

If the cause is not obvious, it is recommended to contact the Grid Control/ SRLDC to obtain information on the situation in the transmission/distribution network outside the switchyard, eg. a transmission system abnormality and its likely duration.

These steps will help not only in eliminating any machine failures, but also minimize the Unit tripping attributable to the excessive NPS currents in the Generator circuit.

Instruction	Responsibility	Document Ref.
Noticing the alarm and identification of the source.	SHIFT OPERATION	<i>LMI/OGN/OPS/ELEC/023</i>
Corrective action as per Step-5.4 above.	SHIFT OPERATION	File 1. DDCMIS/ SER Alarms 2. Shift log book

6.0 METHODOLOGY OF IMPLEMENTATION:

As soon as negative phase sequence, alarm appears, the corrective action as mentioned in step-5.4 above are to be adopted by the shift Operation.

7.0 FREQUENCY OF CHECKING AND MEASUREMENTS:

Tests shall be carried out during unit overhaul.

8.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Basic responsibility lies with Operation Engineer / Desk Engineer. Overall responsibility lies with HOD (O).

9.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION:

Any major abnormality observed, to be reported to Head of O&M.

10.0 RECORD KEEPING:

Whenever any disturbance is observed, copy of alarm page to be taken out by Operation, copy of Generator Currents to be taken from Generator Disturbance Recorder by Electrical Maintenance and submit to EEMG for records.

11.0 REVIEW:

Documents will be reviewed once in two years by HOP.

12.0 CHECKLIST: To be filled by Operation

SL NO	DESCRIPTION	DATE	UNIT NO	REMARKS
1	NPS ALARM STATUS AFTER UNIT SYNCHRONISATION			
2	NPS ALARM APPEARED DURING THE UNIT RUNNING			
3	ALARMS AND CURRENT TRENDS DURING THE TIME OF NPS ALARM			
4	TESTING OF NPS PROTECTION IN RELAY DURING THE ANNUAL OVERHAUL			
5	CHECKING OF NPS ALARM IN MMI DURING THE UNIT OVERHAUL BY SIMULATION			